

Lab Practical #07:

Study the application layer protocol DNS, DHCP, FTP. **Practical**

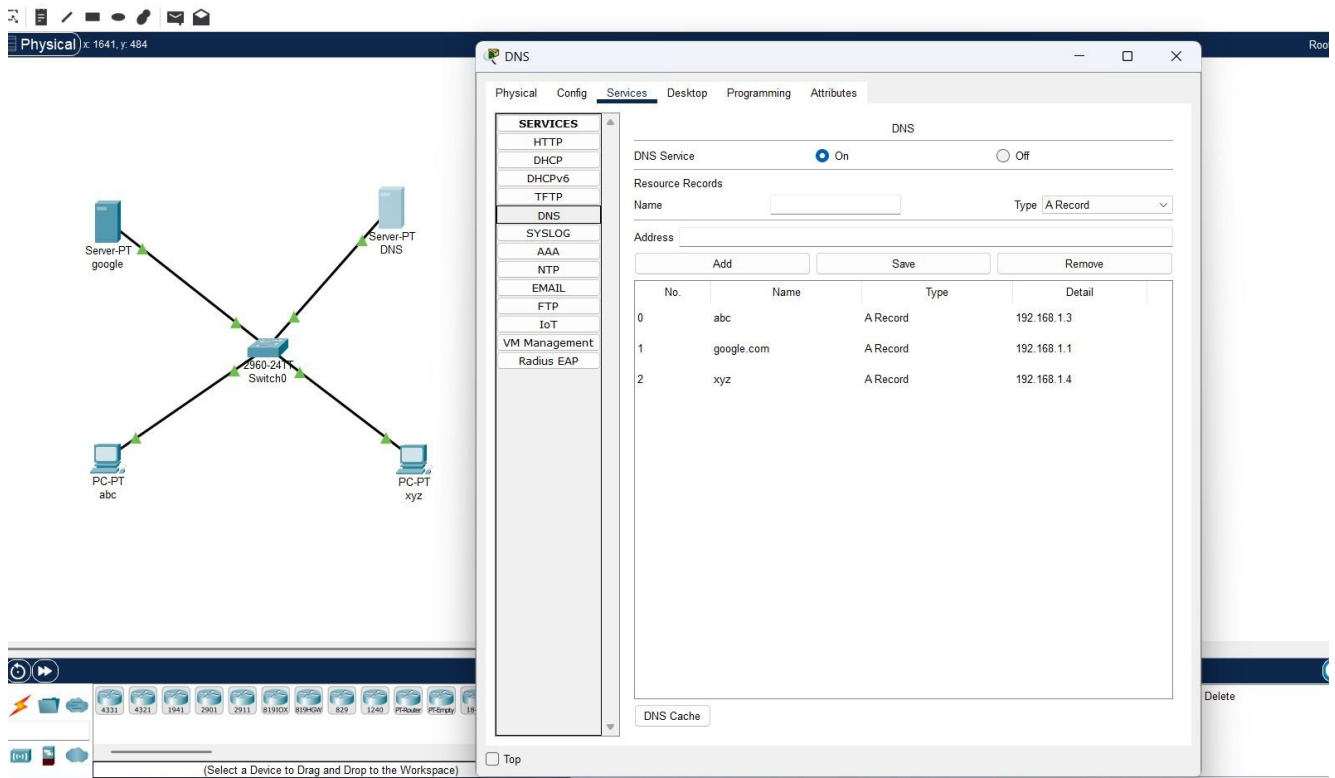
Assignment #07:

1. Implement the application layer protocol DNS, DHCP, and FTP. Also check connectivity between them using ping command or PDU utility.

Instructions:

1. Protocol-wise configuration setup screenshot.
2. Mention IP address of each pc as label.
3. Ping command or PDU screenshot between two pcs.

1. DNS



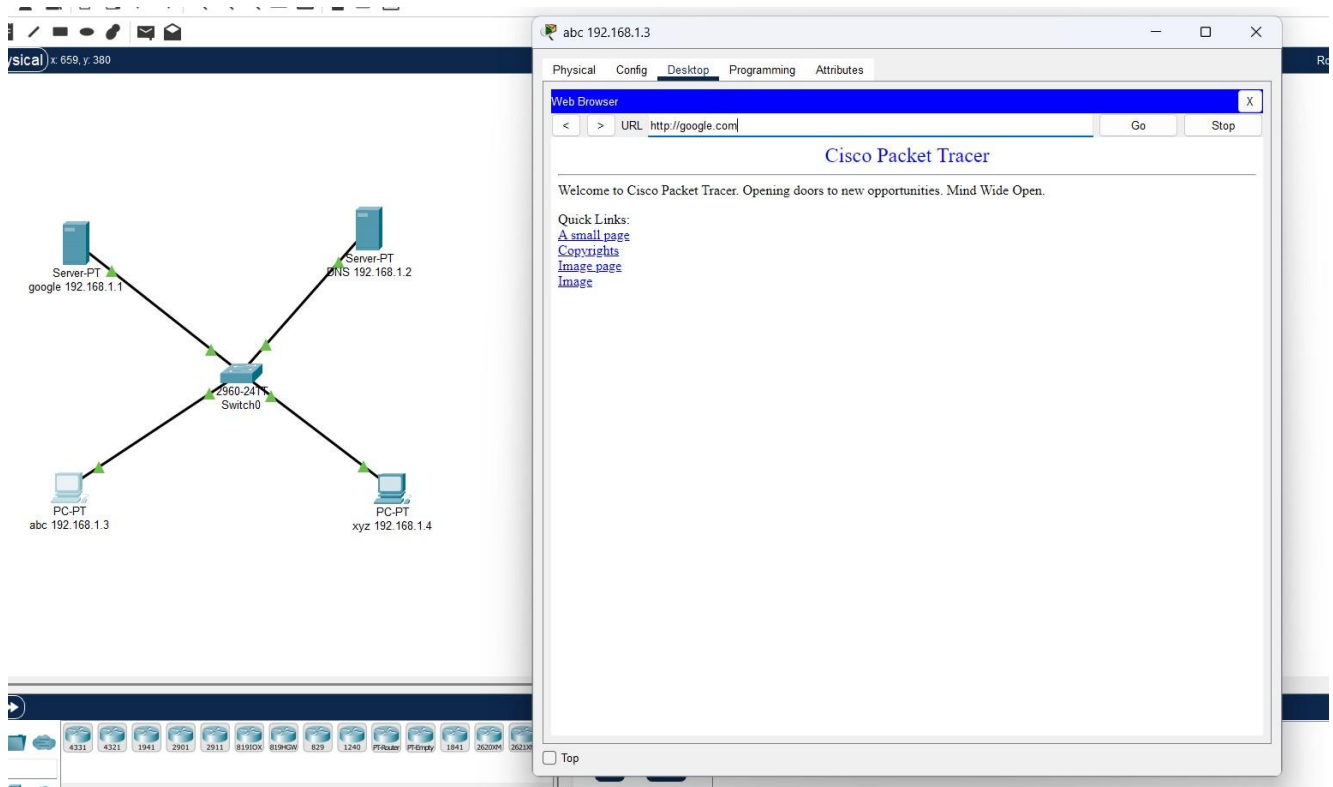
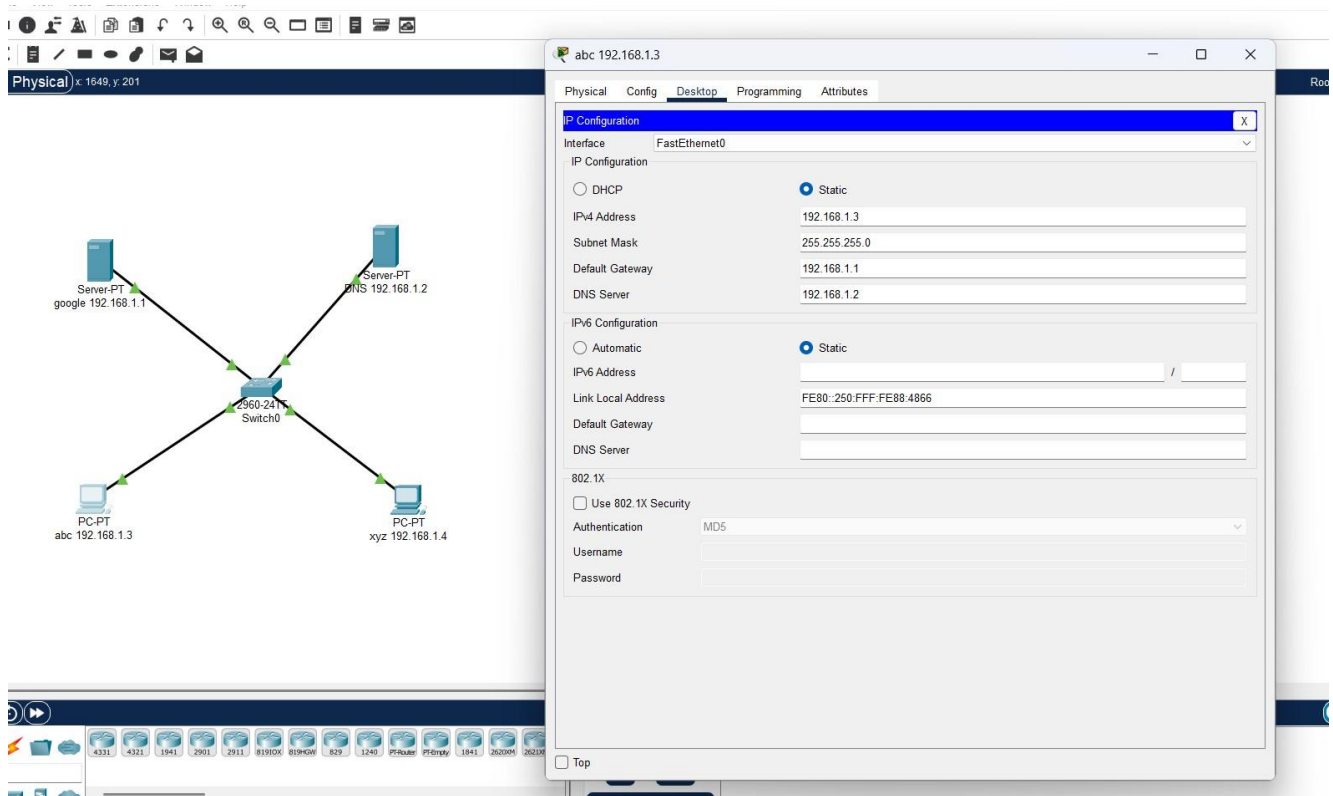
The screenshot displays a network simulation interface. On the left, a topology diagram shows a central '2960-24T Switch0' connected to four devices: 'Server-PT google', 'Server-PT DNS', 'PC-PT abc', and 'PC-PT xyz'. On the right, the 'DNS' configuration window is open, showing the 'Services' tab. The 'DNS Service' is set to 'On'. Below, the 'Resource Records' section contains a table with three entries:

No.	Name	Type	Detail
0	abc	A Record	192.168.1.3
1	google.com	A Record	192.168.1.1
2	xyz	A Record	192.168.1.4

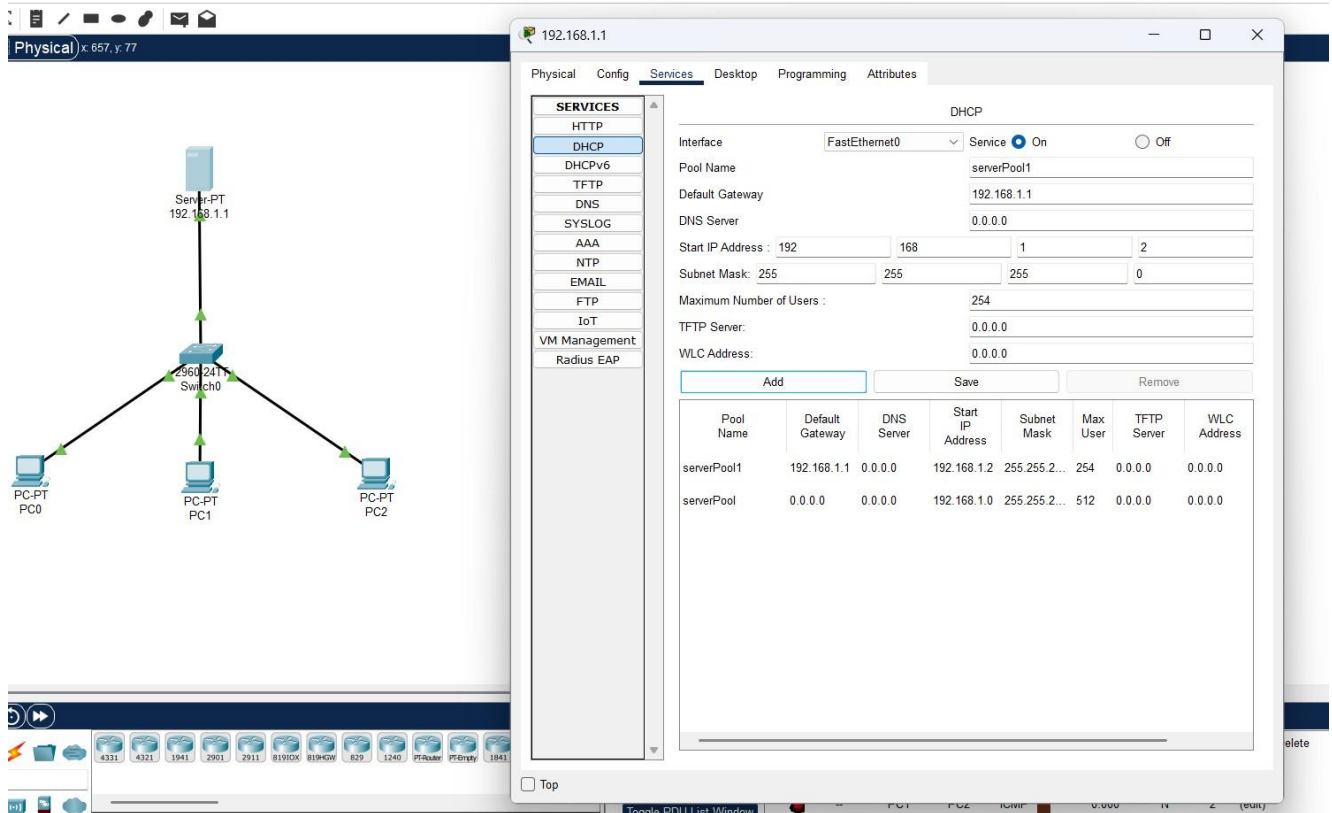
At the bottom of the window, there is a 'DNS Cache' section and a 'Top' button.



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2. DHCP



The screenshot displays a network topology in Packet Tracer. A central 2960 24T Switch0 is connected to three PC-PT devices (PC0, PC1, PC2) and a Server-PT (192.168.1.1). The Server-PT is also connected to the Switch0. The IP address 192.168.1.1 is shown in the top right corner of the window.

The DHCP configuration window is open, showing the following settings:

- Interface: FastEthernet0
- Service: ☒ On
- Pool Name: serverPool1
- Default Gateway: 192.168.1.1
- DNS Server: 0.0.0.0
- Start IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 254
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

The DHCP configuration window also shows a table of DHCP pools:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool1	192.168.1.1	0.0.0.0	192.168.1.2	255.255.255.0	254	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	512	0.0.0.0	0.0.0.0



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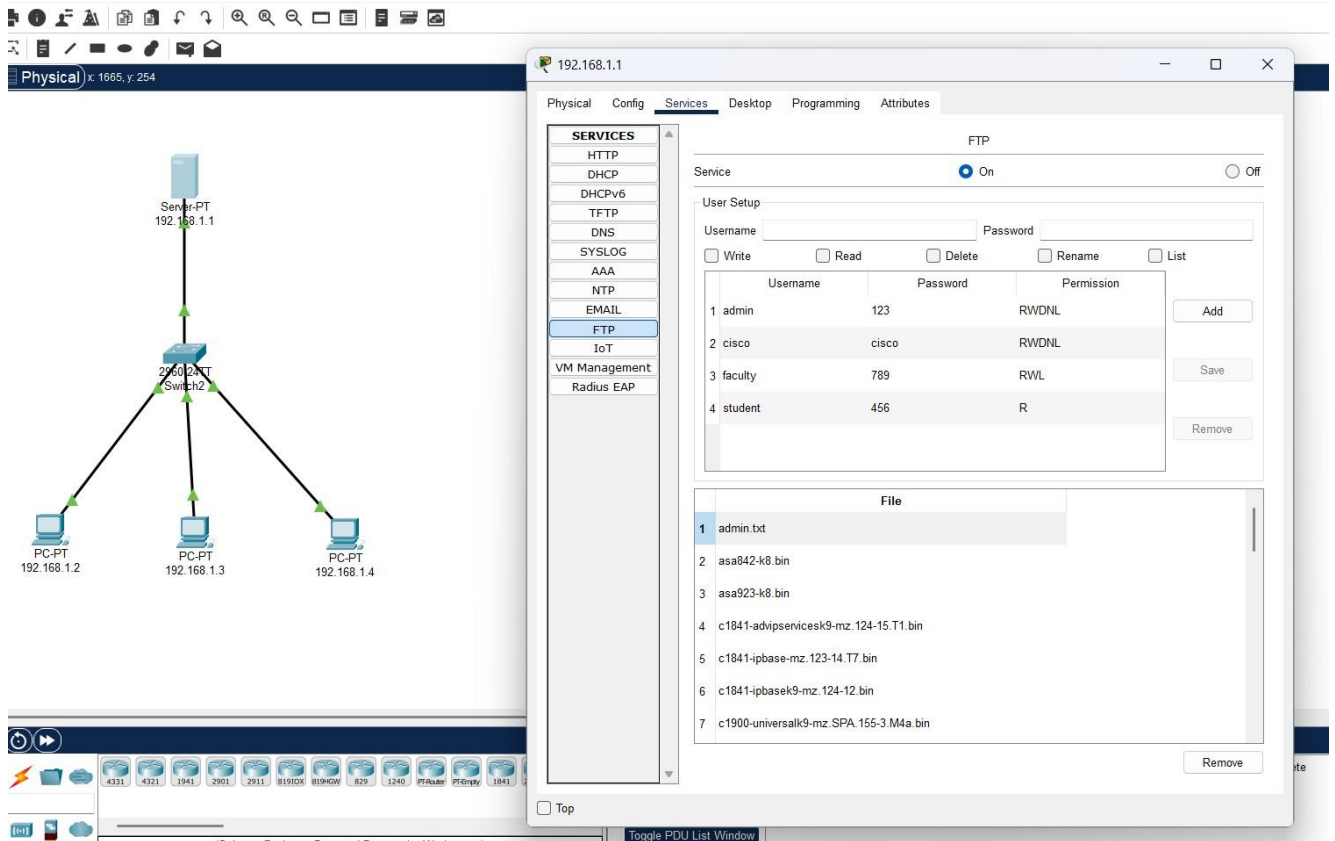
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The network diagram shows a central 2960 24TT Switch0 connected to three PC-PT devices (PC0, PC1, PC2) and a Server-PT (192.168.1.1). The PC0 configuration window is open, showing the IP Configuration tab. The interface is FastEthernet0. The IP Configuration section shows DHCP selected, with IPv4 Address 192.168.1.2, Subnet Mask 255.255.255.0, Default Gateway 0.0.0.0, and DNS Server 0.0.0.0. The IPv6 Configuration section shows Static selected, with IPv6 Address, Link Local Address FE80::20C:CFFF:FE62:6BB0, Default Gateway, and DNS Server. The 802.1X section shows Use 802.1X Security unchecked, Authentication MD5, Username, and Password fields.

The network diagram is identical to the one above. The PC1 configuration window is open, showing the IP Configuration tab. The interface is FastEthernet0. The IP Configuration section shows DHCP selected, with IPv4 Address 192.168.1.3, Subnet Mask 255.255.255.0, Default Gateway 0.0.0.0, and DNS Server 0.0.0.0. The IPv6 Configuration section shows Static selected, with IPv6 Address, Link Local Address FE80::201:63FF:FEC6:6A29, Default Gateway, and DNS Server. The 802.1X section shows Use 802.1X Security unchecked, Authentication MD5, Username, and Password fields.

3. FTP



The screenshot displays a network topology in Packet Tracer. A central 2950-24T switch is connected to a Server-PT (192.168.1.1) and three PC-PTs (192.168.1.2, 192.168.1.3, and 192.168.1.4). The Server-PT is configured with the FTP service. The configuration window for the Server-PT (192.168.1.1) is shown, with the 'Services' tab selected. The 'FTP' service is enabled (On). The 'User Setup' section shows a table of users:

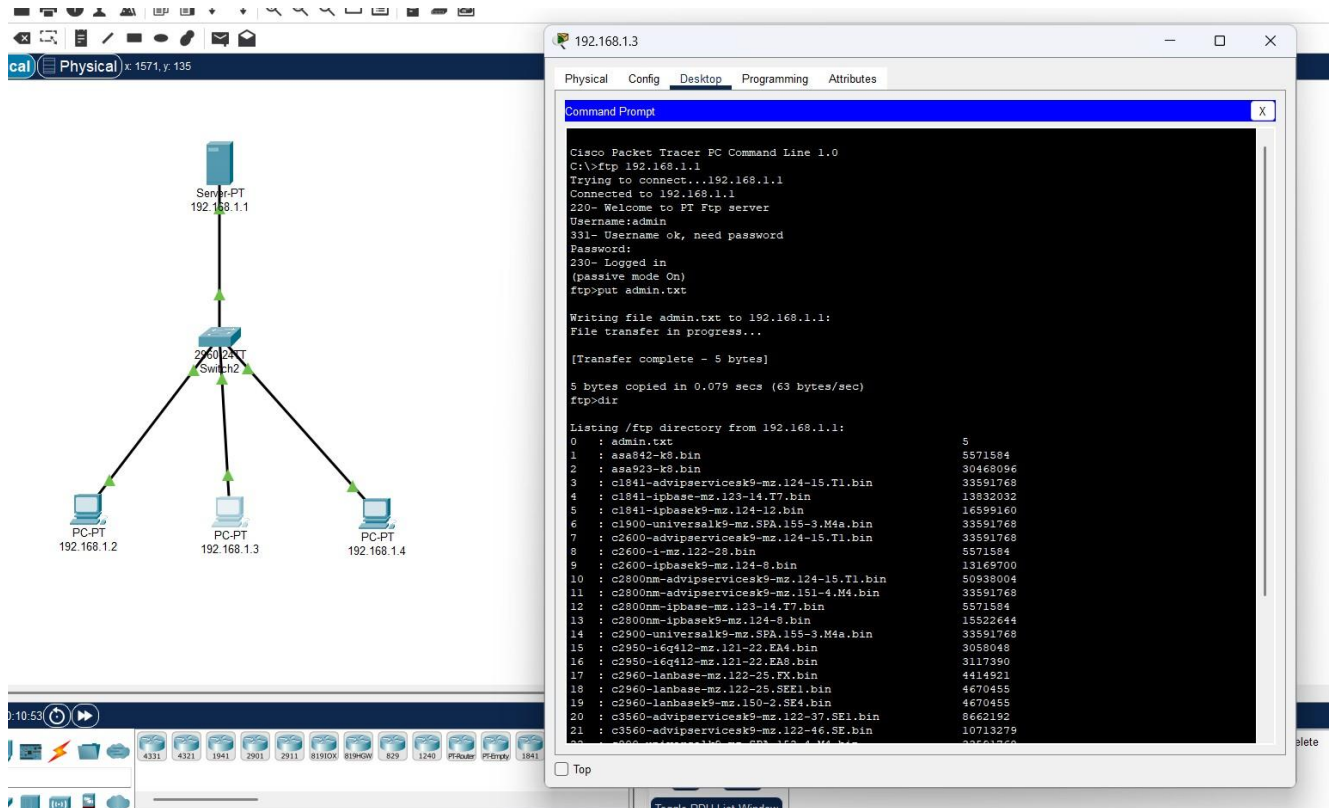
	Username	Password	Permission
1	admin	123	RWDNL
2	cisco	cisco	RWDNL
3	faculty	789	RWL
4	student	456	R

Below the user setup, the 'File' section shows a list of files:

	File
1	admin.txt
2	asa842-k8.bin
3	asa923-k8.bin
4	c1841-adipservicesk9-mz.124-15.T1.bin
5	c1841-ipbase-mz.123-14.T7.bin
6	c1841-ipbasek9-mz.124-12.bin
7	c1900-universalk9-mz.SPA.155-3.M4a.bin



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The network diagram shows a central 2960-24TT Switch2 connected to three PC-PTs (192.168.1.2, 192.168.1.3, 192.168.1.4) and a Server-PT (192.168.1.1). The command window on the Server-PT shows the following output:

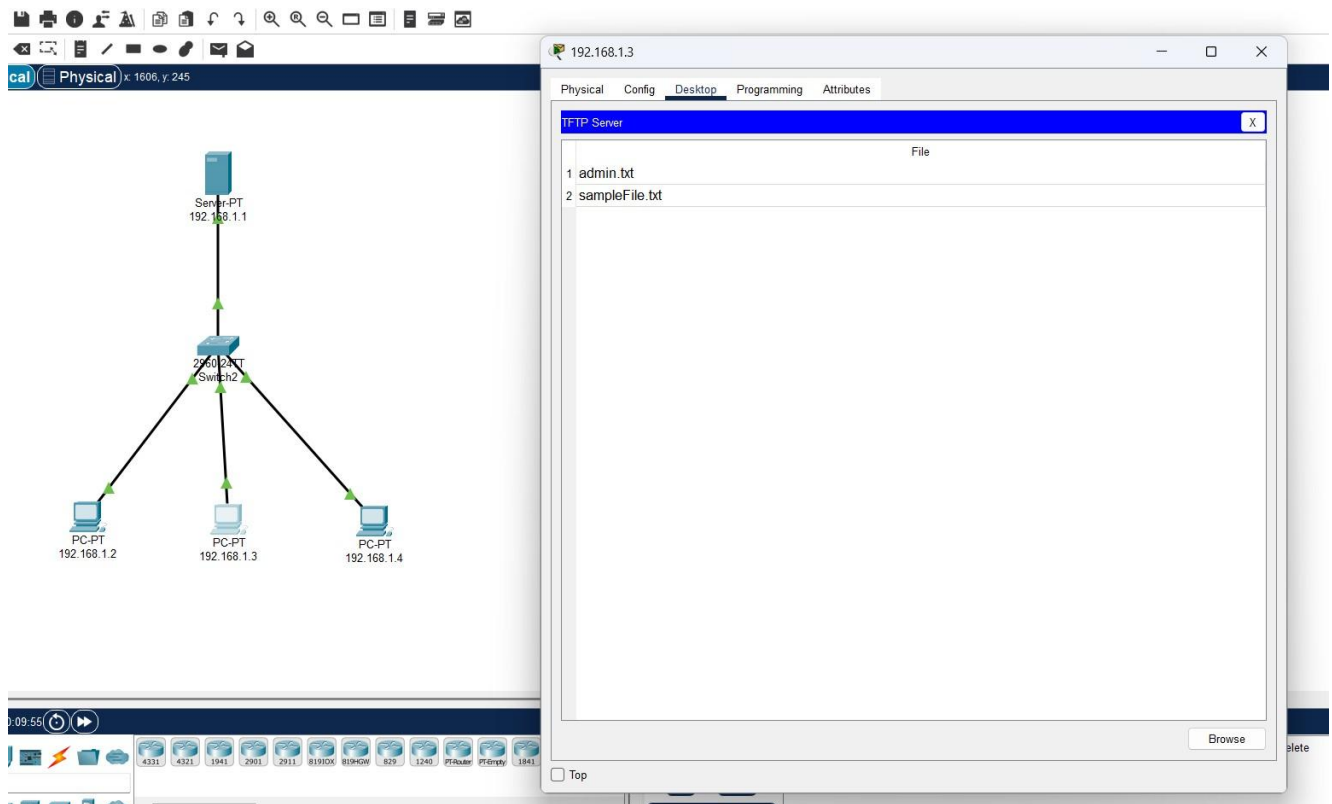
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.1
Trying to connect...192.168.1.1
Connected to 192.168.1.1
220- Welcome to FT Ftp server
Username:admin
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put admin.txt

Writing file admin.txt to 192.168.1.1:
File transfer in progress...

[Transfer complete - 5 bytes]

5 bytes copied in 0.079 secs (63 bytes/sec)
ftp>dir

Listing /ftp directory from 192.168.1.1:
 0 : admin.txt                               5
 1 : asa842-k8.bin                          5571584
 2 : asa923-k8.bin                          30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 4 : c1841-ipbase-mz.123-14.T7.bin          13832032
 5 : c1841-ipbasek9-mz.124-12.bin           16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 8 : c2600-i-mz.122-28.bin                  5571584
 9 : c2600-ipbasek9-mz.124-8.bin            13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
12 : c2800nm-ipbase-mz.123-14.T7.bin        5571584
13 : c2800nm-ipbasek9-mz.124-8.bin          13522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
15 : c2950-16q412-mz.121-22.EA4.bin        3058048
16 : c2950-16q412-mz.121-22.EA8.bin        3117390
17 : c2960-lanbase-mz.122-25.FX.bin         4414921
18 : c2960-lanbase-mz.122-25.SE1.bin        4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin       4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
21 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
22 : c3560-advipservicesk9-mz.151-4.M4.bin 33591768
```



The network diagram is identical to the one above. The command window on the Server-PT shows the output of the 'dir' command:

```
192.168.1.3
Physical Config Desktop Programming Attributes
FTP Server
File
1 admin.txt
2 sampleFile.txt
Browse
```